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PROFESSIONAL SUMMARY

Data Engineer experienced in building scalable data pipelines, cloud-native platforms, and analytics solutions using Kafka, Spark, Airflow, PostgreSQL, and Python. Passionate about driving business value through end-to-end data architectures, from ingestion and transformation to BI dashboards and MLOps deployment.

PROFESSIONAL EXPERIENCE

Data & AI Engineer

February – August 2026

DXC Technology Morocco – Internship

Sale El Jadida, Morocco

- Designed and deployed AI Sandbox, a cloud-native MLOps/AIOps platform for ML benchmarking and AI agent evaluation, containerized with Docker Compose for hybrid on-premise and cloud deployment on Azure VM.
- Built an automated ML benchmarking module (classification & regression) with dataset upload, metric computation, model ranking, and report generation (scikit-learn, XGBoost, LightGBM).
- Developed an AI agent evaluation module with test suites, LLM-as-a-judge scoring, and KPI dashboards for latency, cost, and success rate.
- Implemented a FastAPI backend with PostgreSQL, Redis, MinIO, MLflow, RBAC, Docker Compose, Nginx, and automated Pytest/Vitest tests, delivered in Scrum.
- Technologies: Python, FastAPI, PostgreSQL, Redis, MinIO, MLflow, scikit-learn, XGBoost, LightGBM, Docker, Nginx, Pytest, Vitest, Azure.

Software Engineer & Data Scientist

July – September 2025

JESA S.A. – Internship

Casablanca, Morocco

- Designed CentralGIS, a cloud-native geospatial data management platform with a centralized, containerized architecture (Flask, PostgreSQL/PostGIS, MapLibre GL JS) deployed via Docker.
- Built ingestion pipelines using Kafka, REST APIs, and web scraping to extract and centralize geospatial data from PDFs and reports.
- Reduced data access time by 40% through SQL optimization and PostgreSQL/PostGIS query tuning; ensured consistent cross-environment deployment with Docker containerization.
- Delivered a Data Science project reducing Environmental Impact Assessment inputs from 52 to 38 parameters (27% cut) while maintaining $\geq 95\%$ accuracy via correlated synthetic data generation and XGBoost sentinel selection, replacing a manual Excel macro.
- Technologies: Python, PostgreSQL, PostGIS, Kafka, Docker, Flask, XGBoost, scikit-learn, Pandas.

Full Stack Developer

April – June 2024

OCP Group – Internship

Safi, Morocco

- Developed a digitalized HR management application (Spring Boot, Angular, PostgreSQL) with RFID time tracking, leave requests, absence justification, and automated report generation for managers and employees.
- Built real-time dashboards for workforce analytics (headcount, hours worked, leave, sick days) enabling data-driven decision-making; optimized SQL queries improving backend performance by 30%.
- Collaborated in an Agile team with CI/CD deployment and code reviews on Linux/Docker.
- Technologies: Spring Boot, Angular, PostgreSQL, Docker, CI/CD, SQL.

EDUCATION

National Higher School of Arts and Crafts (ENSAM)

2024 – 2026

Master's Degree in Big Data & Internet of Things – With Honors

Casablanca, Morocco

Faculty of Sciences and Techniques (FST)

2020 – 2024

Bachelor's Degree in Information Systems & Digital Transformation – With Highest Honors

Settat, Morocco

PROJECTS

Real-Time ETL Pipeline – Stock Market

2025

- Built an end-to-end ELT pipeline: Kafka ingestion, PySpark transformation, MongoDB storage, Airflow orchestration with DAG scheduling, PostgreSQL Data Warehouse, and Grafana real-time dashboards, fully containerized with Docker Compose.

Data Quality & KPI Dashboard

2025

- Designed a data quality pipeline: CSV ingestion into PostgreSQL staging, SQL anomaly detection and cleaning (duplicates, NULLs, format inconsistencies), OLAP Data Warehouse loading, and Power BI dashboards with DAX calculated columns tracking completeness, consistency, and data quality KPIs.

- Fine-tuned BERT (Hugging Face Transformers) for multi-label classification of EUR-Lex legal documents. Built a complete ML pipeline: text preprocessing, tokenization, cross-validation, hyperparameter optimization, and multi-class evaluation (precision, recall, F1-score).

SKILLS

- **Programming:** Python, SQL, Java, TypeScript, C, Bash
- **Data Engineering:** Apache Spark, Apache Kafka, Airflow, Hadoop/HDFS, ETL/ELT, Pandas, Selenium, BeautifulSoup
- **Cloud & DevOps:** AWS, Azure, Docker, Docker Compose, Nginx, Git, GitHub Actions, CI/CD, Linux
- **Databases & Warehouses:** PostgreSQL, PostGIS, MongoDB, MySQL, SQL Server, Redis, MinIO
- **Analytics & BI:** Power BI, DAX, Power Query, Grafana, Excel
- **AI & Machine Learning:** scikit-learn, XGBoost, LightGBM, MLflow, BERT, LangChain, NLP, CrewAI, PyTorch, TensorFlow, Prompt Engineering
- **Web Development:** FastAPI, Flask, Spring Boot, React, Angular

CERTIFICATIONS

AWS Cloud Foundations (2026) • Scrum Foundation Learner (2025)

LANGUAGES

- **French:** Fluent **English:** Fluent **Arabic:** Native